DESSERTOPIA

INTRODUCTION :

This report presents SQL queries executed on the Dessertopia E-Commerce Database to analyse order records, sales amounts, and customer spending. Aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() are used to summarise the data and produce useful business information.

Total Number of Orders :

```
SQL> select COUNT(*) FROM ORDERS;

  COUNT(*)
----------
         7
```

Total Sales Amount:

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS ORDERS_TOTAL_AMOUNT FROM ORDERS;

ORDERS_TOTAL_AMOUNT
-------------------
               2800
```

Average Order Amount :

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS ORDERS_TOTAL_AMOUNT FROM ORDERS;

ORDERS_TOTAL_AMOUNT
-------------------
                400
```

Maximum Order Amount :

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS MAXIMUM_AMOUNT FROM ORDERS;

MAXIMUM_AMOUNT
--------------
           550
```

Minimum Order Amount :

```
SQL> SELECT MIN(TOTAL_AMOUNT) AS MINIMUM_AMOUNT FROM ORDERS;

MINIMUM_AMOUNT
--------------
           240
```

Average Amount for Each Order :

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE FROM ORDERS GROUP BY ORDER_ID;

   AVERAGE
----------
       550
       240
       360
       300
       500
       450
       400

7 rows selected.
```

Total Sales Within a Date Range :

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES FROM ORDERS WHERE ORDER_DATE BETWEEN TO_DATE('02-OCT-26', 'DD-MON-RR') AND TO_DATE('06-O
CT-26', 'DD-MON-RR');

TOTAL_SALES
-----------
       1850
```

Activate Windows

Overall Order and Revenue Summary :

```
SQL> SELECT COUNT(ORDER_ID) AS TOTAL_ORDERS,SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE,AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE FROM ORDERS;

TOTAL_ORDERS TOTAL_REVENUE AVERAGE_ORDER_VALUE
------------ ------------- -------------------
           7          2800                 400
```

Total Amount Spent by Each Customer :

```
SQL> SELECT CUSTOMER_ID,SUM(TOTAL_AMOUNT) AS AMOUNT_SPENT FROM ORDERS GROUP BY CUSTOMER_ID;

CUSTOMER_I AMOUNT_SPENT
---------- ------------
C004                300
C006                450
C007                400
C003                360
C005                500
C001                550
C002                240

7 rows selected.
```

Number of Orders Placed by Each Customer :

```
SQL> SELECT COUNT(*) AS ORDER_PLACED_BY_EACH_CUSTOMER FROM ORDERS GROUP BY CUSTOMER_ID;

ORDER_PLACED_BY_EACH_CUSTOMER
-----------------------------
                            1
                            1
                            1
                            1
                            1
                            1
                            1

7 rows selected.
```

Top 5 customers based on total spending :

```
SQL> SELECT *
  2  FROM (
  3      SELECT CUSTOMER_ID,
  4             SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
  5      FROM ORDERS
  6      GROUP BY CUSTOMER_ID
  7      ORDER BY TOTAL_SPENDING DESC
  8  )
  9  WHERE ROWNUM <= 5;

CUSTOMER_I TOTAL_SPENDING
---------- --------------
C001                  550
C005                  500
C006                  450
C007                  400
C003                  360
```

Customers with maximum number of orders :

```
SQL> SELECT CUSTOMER_ID,
  2         COUNT(*) AS NO_OF_ORDERS
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  ORDER BY NO_OF_ORDERS DESC;

CUSTOMER_I NO_OF_ORDERS
---------- ------------
C004                  1
C006                  1
C007                  1
C002                  1
C005                  1
C001                  1
C003                  1
```

High-value customers :

```
SQL> SELECT CUSTOMER_ID,
  2         SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) > (
  6      SELECT AVG(TOTAL_SPENT)
  7      FROM (
  8          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
  9          FROM ORDERS
 10          GROUP BY CUSTOMER_ID
 11      )
 12  )
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I TOTAL_SPENDING
---------- --------------
C001                  550
C005                  500
C006                  450
```

Sales Performance Report :

Total sales :

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS;

TOTAL_SALES
-----------
       2800
```

Number of orders :

```
SQL> SELECT COUNT(*) AS NUMBER_OF_ORDERS
  2  FROM ORDERS;

NUMBER_OF_ORDERS
----------------
               7
```

Average order value :

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
                400
```

Highest order value :

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_VALUE
  2  FROM ORDERS;

HIGHEST_ORDER_VALUE
-------------------
                550
```

Customer Analytics Report :

Customer name, number of orders and total spending:

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  5         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  GROUP BY C.CUSTOMER_ID,
 10           C.FIRST_NAME,
 11           C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I FIRST_NAME                    LAST_NAME            NUMBER_OF_ORDERS
---------- ----------------------------- -------------------- ----------------
TOTAL_SPENDING
--------------
C001       Kavya                         Dharshini                           1
           550

C005       Harini                        Kumar                               1
           500

C006       Divya                         Raj                                 1
           450
```

```
CUSTOMER_I FIRST_NAME                    LAST_NAME            NUMBER_OF_ORDERS
---------- ----------------------------- -------------------- ----------------
TOTAL_SPENDING
--------------
C007       Sneha                         Mohan                               1
           400

C003       Ananya                        Krishnan                            1
           360

C004       Priya                         Sharma                              1
           300


CUSTOMER_I FIRST_NAME                    LAST_NAME            NUMBER_OF_ORDERS
---------- ----------------------------- -------------------- ----------------
TOTAL_SPENDING
--------------
C002       Roshini                       Priya                               1
           240
```